

### LIST OF MOUs

| S.NO.      | NAME OF<br>THE ORGANIZATION/COMPANY/<br>INSTITUTE                      | DATE OF<br>MOU<br>SIGNED |
|------------|------------------------------------------------------------------------|--------------------------|
| 2025-2026* |                                                                        |                          |
| 1          | MONOLITH RESEARCH AND TRAINING LABS PVT LTD                            | 19-09-2025               |
| 2          | WEBZENITH SOLUTIONS                                                    | 19-09-2025               |
| 3          | JEEVISOFTECHNOLOGY PVT LTD                                             | 19-09-2025               |
| 4          | TAMIL NADU CENTRE OF EXCELLENCE FOR ADVANCED<br>MANUFACTURING (TANCAM) | 04-09-2025               |
| 2024-2025  |                                                                        |                          |
| 5          | NSIC- NATIONAL SMALL INDUSTRIES CORPORATION                            | 21-12-2024               |
| 6          | MOVATE TECHNOLOGIES PRIVATE LIMITED                                    | 04-11-2024               |
| 7          | NATIONAL ASSOCIATION OF SOFTWARE AND SERVICE<br>COMPANIES              | 28-10-2024               |
| 8          | DEFSECONE CONSULTING AND TECHNOLOGIES PVT LTD                          | 03-09-2024               |
| 9          | ZOHO CORPORATION PRIVATE LIMITED                                       | 05-08-2024               |
| 10         | AMAZON WEB SERVICES INDIA PRIVATE LIMITED                              | 10-07-2024               |

| 2023-2024 |                                                 |            |
|-----------|-------------------------------------------------|------------|
| 11        | PEARSON INDIA EDUCATION SERVICES PVT.LTD        | 08-09-2023 |
| 12        | PEECEE CASTLE SOFTWARE SERVICES PRIVATE LIMITED | 08-09-2023 |
| 13        | SHL (INDIA) PRIVATE LIMITED                     | 08-09-2023 |
| 14        | ZENARDY TECHNOLOGIES PRIVATE LTD                | 08-09-2023 |
| 15        | FOCUS 4D CAREER EDUCATION PVT LTD               | 08-09-2023 |
| 16        | APTITUDE BUSTER PRIVATE LIMITED                 | 08-09-2023 |
| 2022-2023 |                                                 |            |
| 17        | HEXAWARE TECHNOLOGIES LIMITED                   | 29-05-2023 |
| 18        | TECHGREECH CYBERLABS PRIVATE LIMITED            | 22-05-2023 |
| 19        | FIREFORT TECHNOLOGY SOLUTIONS                   | 09-09-2022 |
| 20        | SILICON SOFTWARE SERVICES                       | 18-08-2022 |
| 21        | VEITECHNOLOGIES PVT.LTD.                        | 03-08-2022 |
| 22        | TRINITY SKILLWORKS PRIVATE LIMITED              | 20-07-2022 |
| 23        | NIDOOS TECHSOLUTIONS PRIVATE LIMITED            | 08-07-2022 |
| 24        | SIDHARTH FOUNDATIONS AND HOUSING LTD            | 17-06-2022 |
| 25        | CELECOM SOLUTIONS GLOBAL PVT LTD                | 25-04-2022 |

### INDUSTRY SUPPORTED LABORATORIES

| S.NO. | NAME OF THE LAB                           | PURPOSE                                                                                                                                                                                                                                    | COMPANY / INDUSTRY PARTNER | TOOLS / TECHNOLOGIES USED                                                                                                       |
|-------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>Internet of Things (Intel)</b>         | IoT Lab is a research platform exploring the potential of crowdsourcing and Internet of Things for multidisciplinary research with more end-user interactions. It empowers students to be part of real-world innovation and social impact. | Intel                      | Intel Galileo / Edison Boards, Arduino, Raspberry Pi, NodeMCU, IoT sensors, Cloud platforms (AWS IoT, ThingSpeak), Python, MQTT |
| 2     | <b>Artificial Intelligence (Idea Lab)</b> | The AI Lab focuses on research in machine learning, robotics, cognitive vision, and NLP. It trains students and facilitates collaboration on automation, robotics, and embedded solutions with hands-on industry-level experience.         | AICTE IDEA Lab / MSME      | Python, TensorFlow, PyTorch, OpenCV, ROS (Robot Operating System), Arduino, Raspberry Pi, MATLAB                                |



| S.NO. | NAME OF THE LAB                             | PURPOSE                                                                                                                                                                                                                                           | COMPANY / INDUSTRY PARTNER            | TOOLS / TECHNOLOGIES USED                                                                    |
|-------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------|
| 3     | AR-VR Goggle Lab                            | AR/VR labs provide hands-on training in designing immersive experiences for gaming, visualization, education, and industrial applications using advanced hardware and software setups.                                                            | Meta / Unity / Unreal Engine Partners | Oculus Rift / Meta Quest, HTC Vive, Unity3D, Unreal Engine, Blender, Autodesk Maya, C#       |
| 4     | Machine Learning Lab (MATLAB)               | This lab supports the research project on <b>Quantum Machine Learning algorithms for traffic management in big cities</b> . Students are trained in AWS Braket and quantum computing fundamentals through weekly collaborative learning sessions. | Arkwith Technologies Pvt. Ltd.        | MATLAB, AWS Braket, Python (NumPy, SciPy, Scikit-learn), Jupyter, Quantum SDKs, GPU Server   |
| 5     | Cyber Security Lab                          | Provides hands-on experience in ethical hacking, network defense, malware analysis, and cryptography. Enables simulation of real-world security threats and research in phishing, threat intelligence, and secure software development.           | DEFSECONE                             | Wireshark, Kali Linux, Metasploit, Burp Suite, Nessus, Nmap, Snort, Splunk, Virtual Machines |
| 6     | Database and Management System Lab (Oracle) | Students analyze case studies to develop business solutions using data modeling and SQL. Focuses on problem-solving, teamwork, and project management while gaining database administration expertise.                                            | Oracle Academy                        | Oracle SQL Developer, MySQL, PL/SQL, MS Access, ER/Studio, Toad for Oracle                   |

| S.NO. | NAME OF THE LAB                                   | PURPOSE                                                                                                                                                                                                                                                            | COMPANY /<br>INDUSTRY<br>PARTNER | TOOLS /<br>TECHNOLOGIES<br>USED                                                                              |
|-------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------|
| 7     | <b>Cloud Computing Lab (VMware, Azure)</b>        | Trains students in virtualization and cloud computing concepts, including VM setup, data center management, and service deployment across cloud platforms.                                                                                                         | VMware and Microsoft Azure       | VMware Workstation/ESXi, Microsoft Azure, AWS, Google Cloud Console, Docker, Kubernetes                      |
| 8     | <b>System Administration Lab (Linux, Windows)</b> | Provides practical training in managing, securing, and maintaining servers. Students learn Linux/Windows commands, user management, patching, and troubleshooting across multi-environment systems.                                                                | Red Hat / Microsoft              | Linux (Ubuntu/CentOS), Windows Server, PowerShell, Bash, Ansible, VMware, Shell Scripting, File System Tools |
| 9     | <b>Networking Lab (Cisco)</b>                     | The Networking Lab enables students to design, configure, and troubleshoot computer networks. It helps them gain expertise in routing, switching, subnetting, and network security. The lab also prepares students for industry certifications like CCNA and CCNP. | Cisco Networking Academy         | Cisco Packet Tracer, Cisco Routers & Switches, GNS3, Wireshark, Network Simulators, VLAN Configuration Tools |





தமிழ்நாடு தமில்நாடு TAMILNADU - 6 SEP 2025



EP 016273  
A. SATHISH KUMAR  
STAMP VENDOR  
LIC NO. 71816183/96  
TRIPLICANE, CHENNAI-600 005  
Phone: 28552348, 43556661

Memorandum of Understanding  
Enterprise Firewall & Security OEM Training & Employment Program

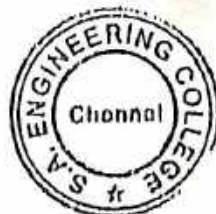
This Memorandum of Understanding (MOU) is executed on this day of 01-Sep-2025.

BETWEEN

S.A Engineering College, Autonomous, duly constituted under S. Amarnaath having its campus and office at Poonamallee -Avadi Road, Veeraraghavapuram, Thiruverkadu, Chennai 600077, hereinafter referred to as the "University / College/First Party", which expression shall where the context so admits includes its successors in interest and permitted assigns.

AND

Movate Technologies Private Limited, a company registered under the Companies Act, 1956 and having its registered office at 6<sup>th</sup> Floor, 32A&B, Ambit IT Park, Ambit IT Park Road, Ambattur Industrial Estate, Chennai - 600058, Tamil Nadu, India, hereinafter referred to as "Movate" / "Second Party", which expression shall where the context so admits includes its successors in interest and permitted assigns.



*S. Amarnaath*

भारतीय गैर न्यायिक

एक सौ रुपये

Rs. 100

₹. 100



ONE  
HUNDRED RUPEES

भारत INDIA  
INDIA NON JUDICIAL

தமிழ்நாடு தமிழ்நாடு TAMILNADU

- 6 SEP 2023



EX-016274  
A. SATHISH KUMAR  
DIRECTOR  
TECHNOLOGICAL EDUCATION  
TAMILNADU

Hereinafter referred to jointly as the "Parties" and individually as a "Party" to this MOU.

WHEREAS

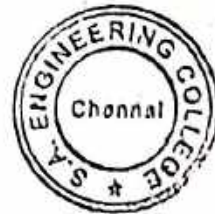
**S.A Engineering College** stands out as an institute of academic excellence by imparting technical, intellectual, and professional skills to the students, offers a distinguishing blend of premium graduate, undergraduate and research programs.

Movate is a global Information Technology, Consulting and Digital Solutions company that provides IT and business consulting, application support and maintenance, development, systems integration, and managed services to its customers.

The College is desirous of placing its students in Movate through Campus Placement - a program conducted within universities and educational institutions to provide jobs to final year passing out students. The educational institution wishes to partner with Movate for recruitment of its final year student population.

Movate is desirous of identifying and nurturing suitable talent from the College who will be considered for placement to Movate after graduation.

This MOU is intended to be legally binding between the Parties and is enforceable in accordance with the applicable laws and regulations.



*Sathish Kumar*



Accordingly, the Parties have decided to reduce into writing their mutual understandings, and understanding as stated hereunder:

NOW THIS MOU WITNESSETH AS FOLLOWS:

Objective:

The overall objective of this MOU is to ensure:

1. Movate partners with University/College to recruit students and conduct the "Enterprise Firewall & Security OEM Training & Employment Program" (the "Program"). Refer Annexure 1
2. Students from the College who meet the eligibility criteria of Movate will be selected and given the final offer by Movate.
3. A Triparty agreement to be signed between the Student, College, and Movate.
4. Selected students to undergo the Program – Technology essentials (80 hrs.) & Technology specialization (120 hrs.) designed to support delivery of Movate's services to its enterprise clients and its affiliates.
5. 200 hours of training to be conducted virtually by Movate spread across 7<sup>th</sup> & 8<sup>th</sup> Semester.
6. Training shall happen turnkey to a maximum of 200 hours (For eg: 4hrs a day 5 days a week; 4hrs a day; 3 days a week)
7. Students will be trained and industry ready to deploy and meet talent availability for Movate.
8. Provide opportunities for students to be groomed and mentored by industry experts.

1. The Parties on this day have executed this MOU to establish a partnership with an intention to enhance the skills of the students selected to Movate from the College through the Program with support from the College SPOC who will be oriented by Movate subject to the following Terms and Conditions:

- a. The College partners with Movate to conduct the Program (Refer Annexure 1).
- b. The Program Sponsors shall be the "Head of Training & Placement Department" from the College and "Head Campus and Institutional Alliance" from Movate.
- c. The Final year students of any Engineering Course with Computer Science/ IT as specialization shall be selected and offered employment by Movate as per Movate's eligibility criteria and selection process and an Offer Letter will be given to the selected Students which will specify the conditions to be complied with to join Movate.
- d. Students will be stack ranked based on their performance from the training program and will be onboarded as per business requirements into Movate roles, subject to the policies of Movate. Movate will have the final say based on the individual performance of the students in this regard.



*Sammuthi*



- e. Movate will provide the selected students with access to the *Accelerate Portal* of Movate with the required e-learning courses for the Program with training content, industry academia exposure, and support events to uplift their capabilities.
  - f. The Program by Movate aims to make the selected Students ready for the Enterprise Product Services Roles at Movate.
  - g. The College shall not charge the selected Students any fee for the course training program and/or recruitment. If it is found that the College has charged fees from students, Movate shall have the right to terminate this MOU and/or pursue other remedies available under the law, against the College.
  - h. Movate would conduct their regular interview process based on their eligibility criteria and the selected students would be given an offer letter, confirming their selection into the Program.
  - i. Movate shall have the discretion to make full-time employment offers to any of the selected students upon successful selection in Movate's recruitment process. Movate shall not be under obligation to recruit any fixed number of Pre-selected Students.
  - j. The Program involves substantial efforts and investment and continuous engagement from Movate with the intent to develop and hire substantial number of students from the College. The College shall, therefore, take all necessary steps to ensure that at least 70% of the pre-selected Students join Movate after successful completion of the training. A drop in the joining percentage below the expected level will seriously jeopardize Movate's interests and may be a material reason to not continue the engagement in the following years.
2. The MOU shall be valid for three (3) years from the date of execution. Movate shall have the right to terminate this MOU for convenience and in the event of breach of material obligations, including confidentiality obligations by the College or students, by giving a notice of fifteen (15) days, provided that the breach is not remedied by the College within such period.
  3. The Parties herein have agreed to perform various obligations for the purpose outlined in the MOU and there is no monetary obligation/consideration involved herein between the Parties. The College shall bear all expenses towards providing infrastructure, network and internet access, other facilities required for education and training and such other expenses incurred in performing the obligations under this MOU.
  4. Under no circumstances shall the Parties be treated as the vendor, employee, contractor, representative and agent of the other. This MOU does not create any other legal relationship of employer/employee, partnership, or joint venture between the Parties herein, other than the relationship contemplated by this MOU. The College does not have any right to obligate or bind Movate in any manner nor the College have the right to act as an agent or representative of Movate.

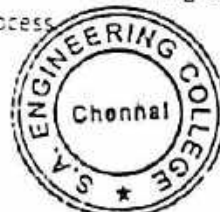


*Amritha*

5. Any information or documentation received by the College during the term of this MOU shall be deemed as "Confidential Information" under this MOU. The Confidential Information includes, but is not limited to, any information, course material, plans, discussions, strategies or any material provided by Movate or its partners to the College and Movate's technology including discoveries, inventions, research and development efforts, manufacturing processes, hardware/software (regardless of media), design and maintenance tools and hardware/software product knowhow, which may be disclosed by Movate to the College during the subsistence of this MOU. Confidential Information shall also include any information marked "Confidential" or "Proprietary" but does not include any information already in the public domain or such information which was obtained from a third party. The College shall not disclose, reproduce, and use any Confidential Information for any purpose except solely in connection with the performance of this MOU. Upon expiration or termination of this MOU, the College shall cease using all such Confidential Information and it shall either destroy or return all copies of such Confidential Information, in whatever form. The College and any of its personnel, students, faculty shall not disclose the Confidential Information during the term of this MOU and thereafter.

The Parties shall not make unauthorized use of the trade name, trademark, copyright, patent, symbol, licenses, or designation belonging to the other party without prior approval and permission. In the event, the College or any of its personnel, students, faculty breach its confidentiality obligations or infringe the intellectual property of Movate, Movate shall have the right to pursue appropriate remedies available at law and shall be entitled to equitable relief. In addition, the College shall indemnify, defend, and hold harmless Movate, from and against all claims and all losses, liabilities, obligations, damages, expenses and costs (including without limitation reasonable legal fees) brought against or suffered by Movate or any of its respective officers, directors, employees, partners or agents, resulting from, arising out of or relating to breach of confidentiality obligations, applicable laws and regulations and infringement of intellectual property.

6. Movate shall provide requisite publicity material such as handouts, information brochures and posters, if applicable to the College. Upon expiration, termination of this MOU or when requested by Movate, the College shall cease using all such material and documents and it shall immediately return all copies of the documents, in whatever form, to Movate.
7. Neither of the Parties shall use the name of the other Party in any advertisement nor make any public announcement without the prior written approval of the other Party.
8. Movate may have similar collaborations with other institutions, and this MOU is not exclusive.
9. The College may have similar collaborations with other companies so long as Movate's interests are not compromised, confidentiality of the knowhow and processes are maintained, the Pre-selected Students alone are put through Movate training curriculum and made available exclusively to Movate for its recruitment process.



*Sammath*



10. The College shall take all necessary steps to safeguard the interest of Movate by avoiding any conflict of interest with other competing companies.

11. Except as expressly stated in this MOU, there shall be no obligation on either of the Parties to compensate the other in any manner or any claim. However, the terms and the conditions in this MOU shall be kept confidential.

12. The Parties agree that no party shall be held liable for any act or omission of the other party under this MOU.

13. The Parties represent that they have the full power and authority to enter this MOU in general and none of the objects stipulated herein are against public policy.

14. The Parties shall designate their respective /representatives who shall be the primary point of contact on behalf of that party as below:

|             | College                                                                          | Movate Technologies Private Limited                                                                     |
|-------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Name        | S. Amarnaath                                                                     | Anish Philip                                                                                            |
| Designation | Correspondent                                                                    | EVP & Chief People Officer                                                                              |
| Email ID    | placement@saec.ac.in                                                             | anish.philip@movate.com                                                                                 |
| Phone       | 9941067999                                                                       | 9742203975                                                                                              |
| Address     | Poonamallee, Avadi Road<br>Veeraraghavapuram, Thiruverkadu,<br>Tamil Nadu 600077 | 6th Floor, 32A&B, Ambit IT Park, Ambattur<br>Park Road, Ambattur Industrial Estate,<br>Chennai - 600058 |

15. The Parties agree to negotiate amendments to this MOU, if necessary, to meet the evolving requirements. Any amendment and / or modifications to the MOU will require written approval from both Parties.

16. The termination of this MOU shall not affect the implementation of the projects or programs established under it, prior to such termination or until such definitive agreement is agreed and executed between the Parties whichever is earlier. Notwithstanding anything to the contrary, in the event of expiry or early termination of this MOU, the College shall reimburse all expenses incurred by Movate in performing its obligations under this MOU till the effective date of termination.

17. Any dispute arising out of this MOU shall be at the first instance attempted to be settled amicably between the Parties within a period of 30 days. If the dispute is not resolved within 30 days from the date it first arose, or if either Party believes that it is unlikely to be resolved in this matter, any such dispute shall be finally referred to arbitration. Any controversy or claim arising out of or relating to this MOU, or the breach thereof, shall be settled in accordance with Indian Arbitration



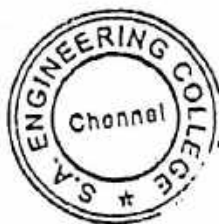
*S. Amarnaath*

and Conciliation Act 1996, and judgment on the award rendered by the arbitrator may be entered in any court having jurisdiction. The arbitration shall be conducted by single arbitrator mutually appointed by the Parties. The seat of arbitration shall be Chennai, Tamil Nadu, India and the arbitration shall be conducted in English language. In case the dispute cannot be settled amicably, this MOU shall be subject to the jurisdiction of courts at Chennai.

18. **Force Majeure.** Neither Party shall be liable for any loss, damage or penalty resulting from delays or failures in performance of their obligations under this MOU if the failure results from events beyond the reasonable control of either Party. For the purpose of this MOU, such events shall include, but are not limited to acts of God, war, hostility, invasion, act of foreign enemies, rebellion, revolution, riots, civil war or disturbances, acts of civil or military authority, laws, regulations, pandemics, epidemics, COVID-19, acts or orders of any governmental authority, body, agency or official, fires, inclement weather, rain or floods (however caused), strikes, lock-outs or other Labour disputes, epidemics, outbreaks, embargoes or other catastrophes affecting the availability of materials or Labour necessary for the performance of this MOU. The Parties hereto agree to notify the other Party promptly of any such circumstances delaying its performance and to resume performance as soon thereafter as is reasonably practicable.
19. **Anti-Bribery Compliance:** The College acknowledges that Movate conducts its business in ethical manner and has no tolerance against bribery and corruption. The College and its promoters, partners, employees, representatives, subcontractors, and agents therefore will not violate any applicable anti-corruption and anti-bribery laws and other laws and regulations in the applicable country of operation, including but not limited to U.S. Foreign Corrupt Practices Act and the Bribery Act 2010, then in force.

IN WITNESS WHEREOF both Parties have caused this MOU executed by their duly authorized representatives on the day, month and year stated here under.

|           |                                                                                     |                                                                                      |
|-----------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Date      | 01 <sup>st</sup> September 2025                                                     | 01 <sup>st</sup> September 2025                                                      |
| Place     | Chennai                                                                             | Chennai                                                                              |
| Name      | S. Amarnaath                                                                        | Anish Philip                                                                         |
| Company   | S.A Engineering College                                                             | Movate Technologies Private Limited                                                  |
| Signature |  |  |





### Annexure 1: Enterprise Firewall & Security OEM Training & Employment Program Proposed

The following is the Enterprise Firewall & Security OEM Training & Employment Program proposed plan to be rolled out by both the Parties for implementation based on agreed timelines.

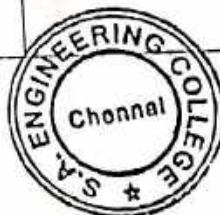
| S. No | Employment & Training Programs                                   | Training Duration in hrs. | Delivery Model          |
|-------|------------------------------------------------------------------|---------------------------|-------------------------|
| 1     | Enterprise Firewall & Security OEM Training & Employment Program | 200                       | E-Learning + VILT + Lab |

### Terms and Conditions for Employment with Movate post completion of Industry Readiness Program

- 100% e-Learning completion and 90% attendance for all VILT sessions
- Passing all training assessments with >80% threshold (up to 2 attempts would be provided for students for each module assessment. If the student fails to clear the first attempt, he should clear the assessment in the second attempt. Failing which, he would not be considered for Movate recruitment process.
- No Backlogs at the time of joining the company.
  - In case the student has obtained a backlog, after joining the program, he/she should ensure completing the same to be eligible for selection to Movate.
  - Once the student clears the backlog, he can apply for opportunities with Movate.
  - Movate would consider employing the student, only if vacancy persists.

### Roles and Responsibilities:

| Movate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | College / University                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Student Orientation &amp; Student Selection</li><li>Conditional Offer Roll out.</li><li>Student Induction &amp; Setting Expectation about the program</li><li>Access to training modules on the portal for the Industry Readiness Program</li><li>Real Time Lab Scenarios</li><li>Master Trainers to deliver master class to students.</li><li>Assessments</li><li>Learning Materials</li><li>Student performance reports would be shared to the college/university every fortnight.</li><li>Certification</li><li>Offer Roll out for successful students.</li><li>Onboarding successful students into Movate</li></ul> | <ul style="list-style-type: none"><li>Sourcing students</li><li>POC to support smooth conduct of training.</li><li>Ensure exclusivity and retention of students selected by Movate for Movate Technologies Private Limited.</li></ul> |



*Sammuthi*

## ENTERPRISE NETWORKING AND SECURITY OPERATIONS

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

### COURSE OBJECTIVES:

- Understand basic networking concepts, models, and addressing.
- Learn firewall architecture and configuration techniques.
- Gain knowledge of VPNs and security protocols.
- Apply troubleshooting methods for products and applications.
- Evaluate service quality and process improvement practices.

**Unit I** 9  
 – Network Fundamentals  
 ISO/OSI Reference Model – IP Addressing – Subnetting – VLSM – CIDR – Ethernet Cabling – Cable Standards and Testing – ARP – RARP – Proxy ARP – GARP – IP Header – Fragmentation – FTP – TFTP – HTTP – HTTPS – DHCP – DNS – TCP/IP Utilities – LAN Routing and Switching : Packet Capture and Analysis using Wireshark – Packet Walk Scenarios – Log Analysis and Interpretation – Troubleshooting Techniques – Layered Approach.

**Unit II** 9  
 – Firewall Technologies  
 Introduction to Firewalls – Positioning in a Network – Stateful vs Stateless Firewall : Session Table vs State Table – State Table Entry Details – Transparent Mode Firewall – Layer 3 Mode Firewall – Zone – Based Firewall – Policy Configuration – Packet Flow through Firewall – TCP, UDP, ICMP, FTP Scenarios – Access Control Lists (ACLs) – Configuration for VPN Access – Network Address Translation (NAT) – Types and Configuration.

**UNIT III** 9  
 Virtual Private Networks and Security Protocols  
 VPN Concepts – Types and Implementation – Understanding and Troubleshooting VPNs - IPsec Protocol – Configuration and Troubleshooting – SSL Messages – Certificate Authority (CA) Certificates – SSL Client Certificate Requests - Secure Communication and Encryption Techniques – Best Practices for VPN and IPsec Deployment.

**UNIT IV** 9  
 Product and Application Troubleshooting  
 Product Architecture – Features and Functionality Overview – Installation – Configuration – Testing Procedures – Product Labs – Problem Simulation and Break Fix Exercises – Packet Capture and Analysis – Tools and Techniques – Debugging Log Files – Log Analysis and Interpretation - Identifying Top Call/Case Drivers – Scenario-Based Discussion and Case Studies – SIP Document and Work Instruction Training.

**UNIT V** 9  
 Quality and Process Management  
 Understanding Service Level Agreements (SLAs) – Key Result Areas (KRA) / Key Performance Indicators (KPI) – Goal Setting – Knowledge Base (KB) Walkthrough and Usage – Quality Guidelines – Process Quality Audits – Case Shadowing – Case Dissection and Analysis – Live Case Handling – Problem Analysis and Break Fix – Feedback and Coaching for Continuous Improvement.

**TOTAL: 45 PERIODS**

### COURSE OUTCOMES:

At the end of the course, the students will be able to



## INDUSTRY SUPPORTED CURRICULUM – MOVATE

### **ENTERPRISE NETWORKING AND SECURITY OPERATIONS**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

#### **COURSE OBJECTIVES:**

- Understand basic networking concepts, models, and addressing.
- Learn firewall architecture and configuration techniques.
- Gain knowledge of VPNs and security protocols.
- Apply troubleshooting methods for products and applications.
- Evaluate service quality and process improvement practices.

#### **UNIT I NETWORK FUNDAMENTALS 9**

ISO/OSI Reference Model – IP Addressing - Subnetting – VLSM – CIDR – Ethernet Cabling – Cable Standards and Testing – ARP – RARP – Proxy ARP – GARP – IP Header – Fragmentation – FTP – TFTP – HTTP – HTTPS – DHCP – DNS – TCP/IP Utilities – LAN Routing and Switching : Packet Capture and Analysis using Wireshark – Packet Walk Scenarios – Log Analysis and Interpretation – Troubleshooting Techniques – Layered Approach.

#### **UNIT II FIREWALL TECHNOLOGIES 9**

Introduction to Firewalls – Positioning in a Network – Stateful vs Stateless Firewall : Session Table vs State Table – State Table Entry Details – Transparent Mode Firewall – Layer 3 Mode Firewall – Zone – Based Firewall – Policy Configuration – Packet Flow through Firewall – TCP, UDP, ICMP, FTP Scenarios – Access Control Lists (ACLs) – Configuration for VPN Access – Network Address Translation (NAT) – Types and Configuration.

#### **UNIT III VIRTUAL PRIVATE NETWORKS AND SECURITY PROTOCOLS 9**

VPN Concepts – Types and Implementation – Understanding and Troubleshooting VPNs - IPsec Protocol – Configuration and Troubleshooting – SSL Messages – Certificate Authority (CA) Certificates – SSL Client Certificate Requests - Secure Communication and Encryption Techniques – Best Practices for VPN and IPsec Deployment.

#### **UNIT IV PRODUCT AND APPLICATION TROUBLESHOOTING 9**

Product Architecture – Features and Functionality Overview – Installation – Configuration – Testing Procedures – Product Labs – Problem Simulation and Break Fix Exercises – Packet Capture and Analysis – Tools and Techniques – Debugging Log Files – Log Analysis and Interpretation - Identifying Top Call/Case Drivers – Scenario-Based Discussion and Case Studies – SIP Document and Work Instruction Training.

#### **UNIT V QUALITY AND PROCESS MANAGEMENT 9**

Understanding Service Level Agreements (SLAs) – Key Result Areas (KRA) / Key Performance Indicators (KPI) – Goal Setting – Knowledge Base (KB) Walkthrough and Usage – Quality Guidelines – Process Quality Audits – Case Shadowing – Case Dissection and Analysis – Live Case Handling – Problem Analysis and Break Fix – Feedback and Coaching for Continuous Improvement.

**TOTAL: 45 PERIODS**

### **COURSE OUTCOMES:**

At the end of the course, the students will be able to

- Explain the fundamentals of networking and troubleshooting.
- Configure and manage firewall operations and policies.
- Implement and troubleshoot VPNs and secure connections.
- Analyze and resolve network and product-related issues.
- Assess and improve service quality and operational processes.



## Tamil Nadu Centre of Excellence for Advanced Manufacturing

Tamil Nadu Industrial Development Corporation (TIDCO) Has Signed A Memorandum Of Understanding (MoU) With Dassault Systems to Establish The 'Tamil Nadu Centre Of Excellence For Advanced Manufacturing' (TANCAM) At the TIDEL Park, Chennai. A First Of Its Kind Centre Of Excellence (CoE) In India. With Its World-Class Team And Performance Excellence For The Industry Segment Like Design, Manufacturing, Manufacturing Execution Systems, Manufacturing Operations Management, High-Fidelity Simulation, Product Design (Both Software & Hardware) And IT Integration For The Entire Enterprise. Headquartered In Chennai, Tamil Nadu Helping The Industry Clusters And MSME To Achieve Their Business Goal And Improve Their Process With Inclusive Sustainability.

TANCAM Focuses On The Area Of Advanced Manufacturing By Leveraging The Strengths Of Partners In The Innovation Ecosystem (Govt, Industry, Academia, MSME, Startups). TANCAM Intends To Collaborate With Industries, Industry Bodies And Academia To Aid In R&D And Skill Development.

TANCAM, The First Of Its Kind, Is A Centre Of Excellence, That Equips Engineering Students With The Necessary Skills And Knowledge To Be Well-Prepared And Successful In The Industry. TANCAM Provides Upskilling Opportunities To Aspiring Students And MSMEs And Focuses On Precision Manufacturing That Requires Upgrading The Skills Of MSMEs. TANCAM Will Facilitate Mentorship And Training Of The Students With Dedicated Domain-Training Courses In Aerospace, Defense, Automotive, Electric Vehicles And Related Industries.

The TANCAM Start-Up Cell Along with Dassault Systems' 3DEXPERIENCE Lab Will Fuel Innovation For Next-Generation Products and New Business Models.

These Are the Following Objectives Of TANCAM:

- Empowering MSME's leveraging Technology
- Supporting Academia through Experimental learning
- Training students and make them Industry ready
- Foster Innovation and support Start-up
- Growth of Aerospace, Defense, Automotive & EV Industries

### Core Tools (on campus)

The foundation of the TANCAM SPOKE Centre is the **3DEXPERIENCE platform** from Dassault Systems, a globally recognised suite of tools. This integrated platform provides comprehensive capabilities across various engineering domains:

- **Design (CATIA):** Create detailed 3D models and product designs for core engineering departments—Mechanical, Civil, Electrical & Electronics (EEE), and Electronics & Communication (ECE).
- **Simulations (SIMULIA – Abaqus):** To perform advanced virtual testing and analysis for all engineering disciplines.
- **Digital Manufacturing (DELMIA):** For planning, simulating, and optimising manufacturing processes.
- **Product Lifecycle Management (PLM) (ENOVIA):** To manage data, collaboration, and the entire lifecycle of a product from conception to disposal.

# INDUSTRY INVOLVEMENT IN THE PROGRAM DESIGN AND CURRICULUM

| S.NO. | NAME OF THE BoS<br>INDUSTRY<br>EXPERT | DESIGNATION               | COMPANY                                            | CONTRIBUTION                  | PERIOD             |
|-------|---------------------------------------|---------------------------|----------------------------------------------------|-------------------------------|--------------------|
| 1.    | Mr.Umashankar<br>Nedunchezian         | CTO                       | Cloud Hub ,<br>Chennai-600015                      | BoS Member<br>Industry expert | 2022-23            |
| 2.    | Dr.R.Gokulakrishnan                   | Additional<br>Director    | Software<br>Technology Parks<br>of India, Chennai. | BoS Member<br>Industry expert | 2023-24<br>2024-25 |
| 3.    | Mr.B.R.SriKiranJyoth                  | Senior Product<br>Manager | PayPal India<br>PvtLtd,Chennai                     | Alumnus                       | 2022-23<br>2024-25 |



**INDUSTRY INVOLVEMENT IN PARTIAL DELIVERY OF ANY REGULAR COURSES  
FOR STUDENTS**

| S.NO. | ACADEMIC<br>YEAR  | NAME OF THE<br>FACULTY      | HOURS<br>TAKEN | MODULE                              |
|-------|-------------------|-----------------------------|----------------|-------------------------------------|
| 1.    | 2024-2025<br>Even | Dr.Shobana                  | 54             | UI/UX Design                        |
| 2.    | 2023-2024<br>Even | Dr.Shobana                  | 54             | Full Stack Development              |
| 3.    | 2022-2023<br>Even | Mr.Rajendran<br>Subramanian | 54             | Oracle, MySQL,<br>MongoDB,AngularJS |

**Visiting Faculty List**



# S.A. ENGINEERING COLLEGE

(Autonomous - Institute Level Research Centre)

Sponsored by Dharma Naidu Educational and Charitable Trust

Accredited by NBA, NAAC with 'A' Grade & ISO 9001 : 2015 Certified Institution

Approved by A.I.C.T.E. and Affiliated to Anna University

Poonamallee - Avadi Main Road, Thiruverkadu, Chennai - 600 077.

Phone : 2680 1999, 2680 1499. Fax : 2680 1899



Visiting

**THIRU.Mr.S. AMARNAATH**  
**CORRESPONDENT**

Date: 13.07.2023

No.SAEC/Rec/2023-2024/Staff/Computer Science and Engineering Department

## APPOINTMENT ORDER

Sub: S.A.Engineering College, Chennai - 77-Ms.P.Shobana Pritha- Appointed  
as Visiting Faculty-**Computer Science and Engineering Department** -  
Orders -Issued..

Ref: Application dated 10.07.2023 of Ms.P.Shobana Pritha

With reference to the application cited above and consequent to the interview attended on 10.07.2019, Ms.P.Shobana Pritha is appointed as **Visiting Faculty** in **Computer Science Engineering Department** of S.A.Engineering College, Chennai. Her Salary will be fixed as per college norms. She will also be eligible to draw other allowances as applicable from time to time.

She will have to submit all the original certificates to the Principal at the time of receiving the Intimation of Selection / Orders of Appointment. No TA/DA will be payable to her to join duty.

She will be on probation for a period of one year from the date of joining duty. She is eligible to draw the Salary as per the college norms during the probation period.

Other terms and conditions of employment will be as per the existing rules of the College. A copy of the conditions of employment is enclosed. She is requested to report for duty to the Principal on **24<sup>th</sup> July 2023**.

Also She is requested to submit his consent letter to the undersigned immediately, in confirmation of the receipt of the orders of appointment. Kindly acknowledge the receipt of the Appointment Order.



*S. Amarnaath*  
**CORRESPONDENT**

To  
Ms.P.Shobana Pritha  
Plot no:02 Phase-II, Garden City Extension  
Varadharajapuram, Manimangalam Road,  
Chennai-63



**S.A. ENGINEERING COLLEGE, CHENNAI - 77.**

**CONDITIONS OF EMPLOYMENT**

01. You are required to carry out all the academic and other duties that may be assigned to you by the Principal of the College and the head of the department.
02. You will not undertake any job or tuition or any other activity in any manner, with or without remuneration, without the consent of the Management.
03. You will strive your best to uphold the name of the institution and aim at promoting the interests and welfare of the institution.
04. You will be eligible for one day casual leave for each month of completed service.
05. After completion of one year of service in the institution you will be eligible for vacation, subject to the condition that all the academic and other duties assigned are carried out before availing the vacation.
06. You will serve for a minimum period of two consecutive semesters i.e. a combination of successive odd and even semesters, or a combination of successive even and odd semesters. You will not be permitted to resign from the post in the middle of the academic year. Also you must serve in this Institution atleast two academic years without break.
07. For resignation during the probationary period, three month's notice in writing or payment of three month's salary in lieu thereof, is mandatory. Three month's notice shall be co-terminus with end of the semester/academic year. All the cases of resignation shall be subject to clause 6 above, as prerequisite.
08. For relieving staff at once due to sick or medical reason the staff member concerned should pay three months salary.
09. You will abide by other rules and regulations specified from time to time by the Principal/Management.
10. The Management reserves the right to terminate the services without notice on breach of discipline or conduct at any time during your service in this institution.
11. The above terms and conditions are subject to revision at the discretion of the Management from time to time.
12. Please sign the enclosed acceptance form for the offer of appointment and conditions of employment and mail the same to the Principal, S.A.Engineering College early.
13. TDS will be deducted from her salary as per rules in force.
14. PF/Medical Insurance/HRA/CCA/Professional Tax and other taxes which are applicable will be deducted from her/his salary periodically at the rate as prescribed by Government of Tamil Nadu.
15. As per the Undertaking given in the Proforma at the time of joining, those who have not completed two years are not eligible for break or prior notice for relieving.
16. One who availed one year break must appear before selection committee as a fresher for continuing in service.



*S. Annamalai*  
CORRESPONDENT

I P. Shobana Pritha have read all these Conditions given above and I agree to obey and serve in this Institution. I accept the offer of appointment as Visiting Faculty in CSE Department with the conditions stipulated by the S.A.ENGINEERING COLLEGE, CHENNAI. I will report for duty on 24/07/2023

DATE: 24/07/23

*P. Shobana*  
APPLICANT SIGNATURE

**S.A. Engineering College, Chennai -77**  
(An Autonomous Institution, Affiliated to Anna University)

**JOINING REPORT**

Date: 24/7/23.

From:

P. Shobana Pritha.

No:02, Phase-II, Garden City Extension,

Varadharaja Puram

Manimangalam Road, Ch-63.

To:

The Principal,  
S.A.Engineering College,  
Chennai- 600 077.

Sub: S.A.Engineering College, Chennai-600 077 – Reporting for duty –Reg.

Ref: Appointment letter from Principal Dated 24/7/23.

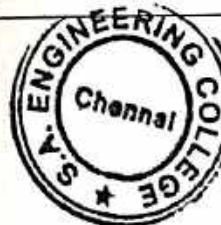
In pursuance of the instructions contained in the appointment letter, I accept the offer of appointment and report for duty as Visiting Faculty in CSE department on 24/7/23 (FN/AN).

I also agree to abide by the conditions of appointment, which may be prescribed.

Thanking you,

Staff Sign:

P. Shobana



PRINCIPAL:  
Dr. S. Ramachandran, M.E. Ph.D.  
PRINCIPAL  
S.A Engineering College (Autonomous).  
Poonamallee-Avadi Main Road,  
Veeraragavapuram,  
Thiruverkadu, Chennai-600 077.



# VEI

TECHNOLOGIES PVT LIMITED

## IDENTITY CARD



EMPLOYEE ID  
VEI 007

  
AB+

**P.SHOBANA PRITHA, MCA., M.Phil. Phd.,**  
**SOFTWARE DEVELOPER & TRAINER**



# **S. A. ENGINEERING COLLEGE, CHENNAI-77**

(Autonomous-Institute Level Research Centre-Affiliated to Anna University)

Accredited by NBA, NAAC with "A" Grade and ISO 9001:2015 certified institution  
Poonamallee – Avadi Main Road, Thiruverkadu, Chennai – 600 077.

## **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**ACADEMIC YEAR: 2025-2026**

**YEAR:III**

### **COURSE SYLLABUS**

#### **FULL STACK DEVELOPMENT**

##### **Course Objectives (COs)**

By the end of this course, students will be able to:

- Understand the fundamentals of web technologies, including HTML, CSS, and JavaScript, and develop responsive and interactive web pages.
- Learn and implement modern front-end frameworks (React) to build dynamic single-page applications with reusable components.
- Develop server-side applications using Node.js and Express.js, design RESTful APIs, and handle client-server communication efficiently.
- Gain proficiency in working with relational and non-relational databases, including CRUD operations, data modelling, and integration with backend servers.
- Integrate front-end, backend, and database into a full-stack application, perform testing and debugging, and deploy projects on cloud platforms.

##### **Module 1: Web Foundations**

11

Introduction to Web Development – Client-Server Architecture, HTTP, APIs-HTML5-Structure of a web Page-Forms, multimedia, semantic tags-CSS3-Selectors, Box model, Flexbox, Grid, Responsive design, animations, Transitions-JavaScript (ES6)-Variables, data types, loops, functions-DOM manipulation and events-JSON and Fetch API,

##### **Module 2: Front-End Framework**

11

Introduction to React (or Angular/Vue – as per institution)-React Components, Props, and State- Hooks (useState, useEffect)- Conditional Rendering and Lists- Event Handling and Forms in React- Routing (React Router)- Integration with REST APIs

##### **Module-3 Backend Development with Node.js**

11

Introduction to Node.js and Express.js- Setting up Express server- Middleware, Routing, and REST API design- Handling requests and responses- Working with JSON and API testing using Postman- Introduction to authentication (JWT / Session)

##### **Module-4 Database and Data Handling**

11

Relational Database (MySQL / PostgreSQL) – CRUD operations- Introduction to NoSQL -MongoDB)- Connecting Node.js with Database (Mongoose / Sequelize)- Schema design and data modelling- Querying and aggregation- Data validation and error handling

##### **Module-5 Integration, Deployment & Project**

10

Integrating Front-end (React) with Backend (Node.js + DB)- Authentication and Authorization- Environment variables and configuration- Deployment – GitHub, Render, Vercel, or AWS- Version Control (Git & GitHub basics)- Testing and Debugging- Project Demo and Evaluation

**Total No.of.Hours: 54**



# Course Outcomes (COs)

After completing the course, students will be able to:

**CO1:** Design and implement responsive, user-friendly web pages using HTML, CSS, and JavaScript.

**CO2:** Build dynamic, component-based single-page applications using React (or chosen front-end framework).

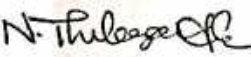
**CO3:** Develop and test server-side RESTful APIs using Node.js and Express.js to handle real-world client requests.

**CO4:** Implement data persistence using relational (MySQL/PostgreSQL) and non-relational (MongoDB) databases and connect them with backend services.

**CO5:** Integrate front-end, backend, and database into a functional full-stack application, and deploy it on cloud platforms while following industry-standard best practices.

| COs | K-Map | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | K3    | 3   | 2   | 2   | 1   | 3   | 1   | 1   | 1   | 2   | 2    | 1    | 2    | 1    |      |
| CO2 | K3    | 3   | 3   |     | 2   | 3   |     | 1   | 1   |     |      |      |      | 1    |      |
| CO3 | K3    | 3   | 3   |     |     |     |     | 1   | 1   |     |      |      |      | 1    |      |
| CO4 | K3    | 3   |     | 3   | 2   |     | 1   |     | 1   | 3   | 2    |      | 2    |      | 1    |
| CO5 | K3    | 3   | 3   | 3   | 2   | 3   |     |     |     |     | 3    | 3    | 3    |      | 1    |

  
FACULTY INCHARGE

  
AEC COORDINATOR

  
HOD/CSE


**S.A. ENGINEERING COLLEGE, CHENNAI-77**  
 (Autonomous-Institute Level Research Centre - Affiliated to Anna University)  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**ACTION TAKEN REPORT BASED ON STUDENT FEEDBACK**

**SUBJECT: FULL STACK DEVELOPMENT**

**ACADEMIC YEAR: 2025-2026**

| Feedback Category                        | Summary of Student Feedback                                                      | Action Taken                                                                                                                                                                                      |
|------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Guest Speaker Knowledge & Presentation   | Majority rated 4.0-5.0, indicating strong delivery and subject expertise.        | Continue inviting domain experts with engaging presentation styles.                                                                                                                               |
| Relevance of Topic                       | High relevance (avg. rating :4.5) to students' academic and career interests.    | Maintain alignment with curriculum and emerging technologies.                                                                                                                                     |
| Q&A Effectiveness                        | Most students felt their queries were addressed (avg. rating: 4.2).              | Encourage interactive Q&A sessions and allocate more time for discussion.                                                                                                                         |
| Engineering Fundamentals (PO1)           | Strong enhancement reported (avg. rating :4.5).                                  | Reinforce foundational concepts through practical examples.                                                                                                                                       |
| Problem Analysis (PO2)                   | Positive impact noted (avg. rating :4.3).                                        | Include more case studies and real-world problem breakdowns.                                                                                                                                      |
| Design & Implementation (PO3)            | Students appreciated design-oriented learning (avg. rating :4.2).                | Introduce mini design challenges during sessions.                                                                                                                                                 |
| Investigation & Data Analysis (PO4)      | Good engagement with analytical tasks (avg. rating :4.3).                        | Provide structured data sets and guided analysis exercises.                                                                                                                                       |
| Use of Modern Tools (PO5)                | High appreciation for tool exposure (avg. rating :4.4).                          | Continue integrating tools like ML, IoT, and cloud platforms.                                                                                                                                     |
| Communication Skills (PO10)              | Moderate improvement noted (avg. rating :4.2).                                   | Include short presentations or peer discussions.                                                                                                                                                  |
| Lifelong Learning Awareness (PO12)       | Strong awareness raised (avg. rating :4.4).                                      | Emphasize evolving tech trends and self-learning resources.                                                                                                                                       |
| Application of Math & Computation (PS01) | Students recognized real-world relevance (avg. rating :4.3).                     | Include more examples linking theory to engineering applications.                                                                                                                                 |
| Domain Knowledge Application (PS02)      | High ratings for practical integration (avg. rating :4.4).                       | Encourage project-based learning with domain-specific tools.                                                                                                                                      |
| Suggestions for Improvement              | Common themes: "Need air conditioner", "Need more sessions", "More informative". | <ul style="list-style-type: none"> <li>- Improve venue comfort (AC installation planned).</li> <li>- Schedule follow-up sessions.</li> <li>- Enhance content depth and interactivity./</li> </ul> |

  
 CLASS ADVISOR

  
 YEAR COORDINATOR

  
 HoD/CSE





**S. A. ENGINEERING COLLEGE, CHENNAI-77**

*(An Autonomous Institution Affiliated to Anna University, Chennai)*

*Accredited by NBA, NAAC with "A" Grade and ISO 9001:2015 certified institution  
Poonamallee – Avadi Main Road, Thiruverkadu, Chennai – 600 077.*

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**ACADEMIC YEAR: 2024-2025**

**YEAR:III**

**COURSE SYLLABUS**

**UI/UX DESIGN**

Course Objectives (COs)

By the end of this course, students will be able to:

- Introduce the fundamental principles of User Interface (UI) and User Experience (UX) design.
- Develop skills in user research, persona creation, and understanding user needs and behavior.
- Enable students to design effective information architecture, wireframes, and prototypes.
- Familiarize students with industry-standard design tools for interface creation and usability testing.
- Empower students to create, evaluate, and refine interactive design systems following usability principles and accessibility standards.

**Module 1: Introduction to UI/UX Design**

**11**

Overview of UI/UX Design-Difference between UI and UX-Design Thinking Process and Human-Centered Design-Human-Computer Interaction (HCI) Basics-Visual Design Principles – Balance, Contrast, Alignment, Hierarchy-Color Theory, Typography, and Layout Design-Case Study: Evaluating existing website interfaces

**Module 2: User Research and Requirement Analysis**

**11**

Introduction to User Research-Types of User Research: Qualitative and Quantitative-Data Collection Methods: Interviews, Surveys, Observations-Empathy Mapping and Persona Creation-User Journey Mapping and Scenario Building-Defining Design Goals and Problem Statements

**Module 3: Information Architecture and Wireframing**

**11**

Information Architecture – Concepts and Importance-Content Hierarchy and Navigation Design-Creating Sitemaps and Flowcharts-Wireframing Techniques – Low-fidelity & High-fidelity-Layout Design and Visual Hierarchy-Tools for Wireframing (Figma, Adobe XD, Sketch)

**Module 4: Prototyping and Visual Design**

**11**

Introduction to Prototyping-Interactive Components and Transitions-Style Guides, Design Systems, and UI Kits-Color Schemes, Typography, and Iconography-Responsive Design Principles (Mobile, Tablet, Desktop)-Accessibility and Inclusive Design

**Module 5: Usability Testing and Project Development**

**10**

Usability Testing Concepts and Methods-Heuristic Evaluation and Cognitive Walkthrough-Accessibility Standards (WCAG)-Collecting and Analysing User Feedback-Design Iteration and Refinement-Final Project Presentation and Review

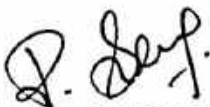
**Total No.of.Hours: 54**

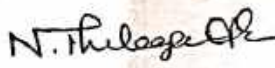
# Course Outcomes (COs)

After completing the course, students will be able to:

- CO1:** Explain the principles and process of UI/UX design and its role in digital product development.
- CO2:** Conduct user research and create user personas, empathy maps, and journey maps to understand user requirements.
- CO3:** Design user interfaces through wireframes, information architecture, and prototypes using standard design practices.
- CO4:** Apply usability testing techniques to evaluate and improve interface designs using tools such as Figma, Adobe XD, or Sketch.
- CO5:** Integrate aesthetics, accessibility, and interaction design principles to create responsive and user-centered digital experiences.

| COs | K-Map | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | K2    | 3   | 2   | 2   | 1   | 3   | 1   | 1   | 1   | 2   | 1    | 1    | 2    | 1    |      |
| CO2 | K3    | 3   | 2   |     |     |     |     |     |     | 2   |      |      |      | 1    |      |
| CO3 | K3    | 3   | 2   |     |     |     |     |     |     | 2   |      |      |      | 1    |      |
| CO4 | K3    | 3   | 1   | 3   | 2   |     | 1   |     |     | 2   |      |      | 2    |      | 1    |
| CO5 | K4    | 3   | 3   | 3   | 2   | 3   |     |     |     | 2   | 2    | 2    | 2    |      | 1    |

  
FACULTY INCHARGE

  
AEC COORDINATOR

  
HoD/CSE





**S.A. ENGINEERING COLLEGE, CHENNAI-77**  
(Autonomous-Institute Level Research Centre - Affiliated to Anna University)  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**ACTION TAKEN REPORT BASED ON STUDENT FEEDBACK**

**SUBJECT: UI/UX DESIGN**

**ACADEMIC YEAR: 2024-2025**

| Content                             | Student Feed back                                                                                                                                                                                                                                                                                                                                                                                                 | Action Taken                                                                                                                                                                                                                    |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.Content and Subject Relevance     | Content Delivery response rating: 4.32/5<br>Subject Relevance response rating: 4.8/5<br>Students appreciated that the visiting faculty covered topics aligned with current industry trends and their academic curriculum.                                                                                                                                                                                         | Future sessions will continue to emphasize domain-aligned content, integrating emerging technologies such as AI, IoT, and cloud computing.                                                                                      |
| 2. Delivery and Presentation Style  | Students rated the knowledge level and presentation style of the guest speakers at 4.46/5, indicating high satisfaction with the delivery and clarity.                                                                                                                                                                                                                                                            | Guest lecturers will be encouraged to continue using interactive delivery methods—such as live demonstrations and Q&A segments                                                                                                  |
| 3. Query Resolution and Interaction | The Q&A sessions received a rating of 4.32/5, showing that students' queries were addressed effectively. However, a few responses suggested limited time for discussions.                                                                                                                                                                                                                                         | Future events will allocate extended Q&A time and include a post-session discussion forum or feedback corner for additional clarifications                                                                                      |
| 4. Learning Outcomes (POs and PSOs) | Mapped PO's<br>1. Engineering knowledge (PO1) with an average score of 4.19.<br>2. Problem analysis skills (PO2) with an average score of 4.13.<br>3. Design and implementation aspects (PO3) scored 4.15<br>4. Lifelong learning (PO12) with an average score of 3.85<br>Mapped PSO's<br>1. Mathematical and computational concepts (PSO1) was rated 4.36<br>2. Domain knowledge integration (PSO2) scored 4.48. | Future events will include hands-on problem-solving activities and mini project demonstrations to strengthen design and implementation skills.                                                                                  |
| 5. Logistics and Environment        | Some students suggested improvements such as "Need Air Conditioner" and "Need some more sessions", indicating logistical concerns and interest in longer or additional sessions.                                                                                                                                                                                                                                  | Arrangements have been made to improve classroom ventilation and comfort during sessions.                                                                                                                                       |
| 6. Overall Impression               | Most students responded with positive comments such as " <i>Very good session</i> " and " <i>More informative</i> ", indicating high satisfaction with the visiting faculty program.                                                                                                                                                                                                                              | The department acknowledges the encouraging feedback and plans to maintain this quality by inviting diverse experts from academia and industry, ensuring that sessions remain informative, engaging, and technically enriching. |

CLASS ADVISOR

YEAR COORDINATOR

HOD/CSE



# S.A. ENGINEERING COLLEGE

(AUTONOMOUS)

Sponsored by Dharma Naidu Educational and Charitable Trust  
Accredited by NBA, NAAC with 'A' Grade & ISO 9001 : 2015 Certified Institution  
Approved by A.I.C.T.E. and Affiliated to Anna University  
Poonamallee - Avadi Main Road, Thiruverkadu, Chennai - 600 077.  
Phone : 2680 1999, 2680 1499, Fax : 2680 1899



THIRU.D.DASARATHAN.,  
SECRETARY

Date: 10.07.2019

No.SAEC/Rec/19-20/Staff/CSE

## APPOINTMENT ORDER

Sub: S.A.Engineering College, Chennai - 77-Mr.S.Rajendran- Appointed as  
Visiting Faculty-CSE Department - Orders -Issued.

Ref: Application dated 10.07.2019 of Mr.S.Rajendran

With reference to the application cited above and consequent to the interview attended on 10.07.2019, **Mr.S.Rajendran** is appointed as **Visiting Faculty** in **Computer Science Engineering Department** of S.A.Engineering College, Chennai. His Salary will be fixed as per college norms. He will also be eligible to draw other allowances as applicable from time to time.

He will have to submit all the original certificates to the Principal at the time of receiving the Intimation of Selection / Orders of Appointment. No TA/DA will be payable to him to join duty.

He will be on probation for a period of one year from the date of joining duty. He is eligible to draw the Salary as per the college norms during the probation period.

Other terms and conditions of employment will be as per the existing rules of the College. A copy of the conditions of employment is enclosed.

He is requested to report for duty to the Principal on **15<sup>th</sup> July 2019**.

Also He is requested to submit his consent letter to the undersigned immediately, in confirmation of the receipt of the orders of appointment. Kindly acknowledge the receipt of the Appointment Order.

  
SECRETARY

To  
Mr.S.Rajendran,  
Plot No77, Swami Vivekananda Nagar,  
Surveyor Colony,  
Madurai -625007





## S. A. ENGINEERING COLLEGE, CHENNAI-77

(An Autonomous Institution Affiliated to Anna University, Chennai)

Accredited by NBA, NAAC with "A" Grade and ISO 9001:2015 certified institution

Poonamallee – Avadi Main Road, Thiruverkadu, Chennai – 600 077.

### DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC YEAR: 2022-2023

YEAR:III

#### COURSE SYLLABUS

#### Oracle (Mysql) and Mongo DB Programming with Angular JS

##### Course Objectives

The course aims to:

- Introduce students to relational database management using Oracle/MySQL and non-relational (NoSQL) database concepts using MongoDB.
- Enable learners to perform database design, schema creation, and data manipulation using SQL and PL/SQL.
- Provide hands-on experience with MongoDB CRUD operations, aggregation, and indexing.
- Develop skills in AngularJS for creating dynamic and responsive front-end applications.
- Integrate front-end (AngularJS) with backend databases (MySQL and MongoDB) to build complete data-driven web applications.

##### Module 1: Relational Database Concepts and MySQL/Oracle Fundamentals

11

Introduction to RDBMS concepts – Data, Tables, Relations, Keys-SQL Basics – DDL, DML, DCL, TCL commands-Creating and modifying databases, tables, and constraints-Queries: SELECT, WHERE, ORDER BY, GROUP BY, HAVING-Joins, Subqueries, and Views-Indexing and normalization (1NF–3NF)-Introduction to Oracle & MySQL Workbench

##### Module 2: Advanced SQL and PL/SQL Programming

11

Introduction to PL/SQL Architecture-Variables, Data Types, and Control Structures-Cursors, Triggers, and Stored Procedures-Exception Handling in PL/SQL-Functions and Packages in Oracle/MySQL-Transactions and Error Handling-Performance Optimization and Query Tuning

##### Module 3: NoSQL Database – MongoDB Programming

11

Introduction to NoSQL and MongoDB Architecture-Difference between SQL and NoSQL Databases MongoDB Data Model – Collections and Documents-CRUD Operations (Insert, Find, Update, Delete)-Query Operators, Projections, and Sorting-Indexing, Aggregation, and MapReduce-Connecting MongoDB with Node.js or Python

##### Module 4: AngularJS Fundamentals

11

Introduction to JavaScript MVC Frameworks and AngularJS Overview-AngularJS Architecture – Modules, Controllers, and Directives-Data Binding (One-way and Two-way)-Filters and Expressions-Dependency Injection and Services-Form Validation and Routing-Connecting AngularJS to REST APIs (HTTP Services)

##### Module 5: Integration and Project Development

10

Integrating AngularJS Front-End with MySQL / MongoDB Back-End-Building REST APIs using Node.js / Express for Data Access-JSON Data Interchange between Front-End and Database-Authentication and Error Handling-Deployment using GitHub / Render / Vercel-Final Project Presentation and Evaluation

Total No.of.Hours: 30

### Course Outcomes (COs)

After completing this course, students will be able to:

- CO1:** Apply the fundamentals of relational and non-relational database systems.
- CO2:** Design and implement database schemas, constraints, and queries using Oracle/MySQL.
- CO3:** Perform data operations and aggregation in MongoDB for unstructured data management.
- CO4:** Develop dynamic web applications using AngularJS components, directives, and services.
- CO5:** Integrate AngularJS front-end applications with MySQL/MongoDB backends to create real-world data-driven projects.

| COs | K-Map | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-----|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | K3    | 3   | 2   | 2   | 1   | 3   | 1   | 1   | 1   | 2   | 2    | 1    | 2    | 1    |      |
| CO2 | K3    | 3   | 3   | 1   | 2   | 3   | 1   | 1   | 1   |     | 1    |      |      | 1    |      |
| CO3 | K3    | 3   | 3   | 1   | 1   |     | 1   | 1   | 1   |     | 1    |      |      | 1    |      |
| CO4 | K3    | 3   | 2   | 3   | 2   |     | 1   |     | 1   | 3   | 2    |      | 2    |      | 1    |
| CO5 | K3    | 3   | 3   | 3   | 2   | 3   | 1   |     |     |     | 3    | 3    | 3    |      | 1    |

  
FACULTY INCHARGE

AEC COORDINATOR

HoD/CSE